ETH Zürich
Department of Computer Science

“Optimistic Active Exploration of Dynamical Systems” (Sukhija et al., NeurIPS 2023)

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Foundations of Reinforcement Learning (263-5255-00L), Spring 2025

Outline

1. Introduction
2. Problem Formulation
3. Related Work and Contributions
4. Methodology: Optimism in Information Space
5. Theoretical Guarantees
6. Empirical Evaluation
7. Discussion & Conclusion

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Motivation: Beyond Task-Specific Models ben

Current Model-Based RL:

- Learn dynamics f for **one specific reward**
- Model accurate only where reward is high
- Cannot generalize to new tasks

Problem

Task-specific exploration

$\Downarrow$

Biased dynamics models

$\Downarrow$

Poor generalization

What We Want:

- Learn **task-agnostic** dynamics model
- Accurate **everywhere** in reachable space
- Zero-shot transfer to any reward

Key Insight: Explore to reduce **epistemic uncertainty** about dynamics, not to maximize reward

Research Question

How can we efficiently explore to learn accurate dynamics models that generalize to *any* downstream task?

Introducing OPAX

- **OPAX:** Optimistic Active Exploration of Dynamical Systems.[1]
- **Core Idea:** Frame exploration as Bayesian optimal experimental design.
  - Actively seek interactions (trajectories) that maximally reduce *epistemic uncertainty* about the unknown dynamics f^*.
- **Key Components (Overview):**
  1. **Probabilistic Model:** Quantifies belief and uncertainty about f^* (e.g., Gaussian Process, Ensemble).
  2. **Information Gain Objective:** Formalizes uncertainty reduction as the goal.
  3. **Optimistic Planning:** Plans trajectories assuming the "most informative" plausible dynamics.

Promise of OPAX

Learn general-purpose world models efficiently, enabling robust planning and control across diverse tasks without task-specific retraining.

[1] **sukhija2023optimistic**.

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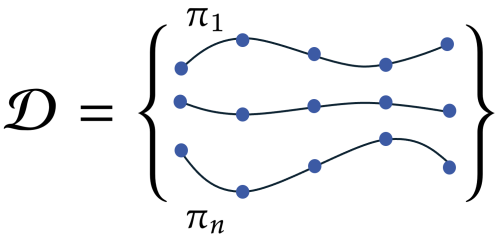
Problem Formulation (1/2)

Given an unknown discrete-time dynamical system $f^\star : \mathcal{S} \times \mathcal{A} \to \mathcal{S} \subseteq \mathbb{R}^d$:

$$s_{t+1} = f^\star(s_t, a_t) + w_t, \quad w_t \sim \mathcal{N}(0, I).$$

Goal: Design algorithms for learning $\hat{f} \approx f^\star$ that is good for any reward functions in episodic setting:



Introduction video by Bhavya Sukhija. NeurIPS 2023.

Problem Formulation (2/2)

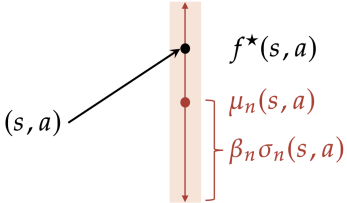
Our problem can be divided into two parts:

- Exploration (our focus!): Collecting data by executing policies.
- Estimation: Building model from data.

Well-Calibration Assumption

Given n data, we can construct confidence intervals:

$$|f_i^\star(s,a) - \mu_{n,i}(s,a)| \leqslant \beta_n \sigma_{n,i}(s,a) \quad \forall i \in [d], \text{ w.h.p.},$$



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Related Work: Active Learning

	Bandit Optimization	Our Problem
Goal	$\max_x f^\star(x)$	Find $\hat{f} \approx f^\star$
$f^\star$	Objective	Dynamics
State space	No	Continuous
Planning	No	Yes
Technique	Optimism in the face of uncertainty	

N. Srinivas et al. "Gaussian process optimization in the bandit setting: No regret and experimental design." ICML 2010. (The 1st presentation today!)

Related Work: Maximum Entropy RL

What about collecting data by maximum entropy or entropy-regularized RL?

$$\max_{\pi} H\left(d_{\mu}^{\pi}\right), \quad \max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^{t}(r(s_t, a_t) - \beta \log \pi(a_t|s_t)) \,\middle|\, s_0 \sim \mu, \pi\right].$$

- No guarantee for arbitrary reward functions in downstream tasks.

E. Hazan et al. "Provably efficient maximum entropy exploration." ICML 2019. (The 3rd presentation today!)
B. Eysenbach and S. Levine. "Maximum entropy RL (provably) solves some robust RL problems." ICLR 2022.

Related Work: Reward-Free RL

- Study this problem in tabular settings or impose structural assumptions on values function and transitions.
- No practical algorithms.

C. Jin et al. “Reward-free exploration for reinforcement learning.” ICML 2020.
R. Wang et al. “On reward-free reinforcement learning with linear function approximation.” NeurIPS 2020.
J. Chen et al. “On the statistical efficiency of reward-free exploration in non-linear RL.” NeurIPS 2022.

Contributions

- Propose OpAX, a practical algorithm for learning non-linear dynamics with continuous state-action spaces and without structural assumptions.
- In this setting, OpAX is the first algorithm to have a theoretical guarantee for a rich family of nonlinear dynamics.

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How OPAX Works: Information as Reward ben

Key Idea: Transform exploration into a sequential decision problem with intrinsic rewards

1. Model Uncertainty

- Learn dynamics with uncertainty: (μ_n, σ_n)
- GP or Neural Network Ensemble

2. Information Reward

$$r_{\text{info}}(s, a) = \sum_j \log\left(1 + \frac{\sigma^2_{n,j}(s, a)}{\sigma^2_w}\right)$$

High reward = high uncertainty

3. Exploration Strategy

Traditional RL

Maximize task reward

$\Downarrow$

Biased exploration

OPAX

Maximize **information gain**

$\Downarrow$

Comprehensive exploration

Why Information Gain?

- Principled: Rooted in Bayesian experimental design
- Task-agnostic: No bias towards specific rewards
- Diminishing returns: Log ensures convergence

Step 1: Information-Theoretic Objective ben

Ideal Goal: Maximize mutual information between dynamics f^* and collected data

Greedy Episodic Approximation: At episode n, choose policy to maximize expected information gain:

$$\pi_n^* = \arg\max_{\pi \in \Pi} \mathbb{E}_{\tau^\pi} \left[I \left(f^*_{\tau^\pi}; y_{\tau^\pi} \mid \mathcal{D}_{1:n-1} \right) \right] \tag{1}$$

where:

- $f^*_{\tau^\pi} = (f^*(z_0), \ldots, f^*(z_{T-1}))$ - true dynamics along trajectory
- $y_{\tau^\pi} = (y_0, \ldots, y_{T-1})$ - observed next states
- $\mathcal{D}_{1:n-1}$ - data from previous episodes

Challenge

This objective is intractable in general. Need tractable approximation.

Step 2: Tractable Upper Bound ben

Lemma (Information gain upper bound)

For Gaussian noise $w \sim \mathcal{N}(0, \sigma^2 I)$ and epistemic uncertainty σ_{n-1}:

$$I(f^*_{\tau^\pi}; y_{\tau^\pi} \mid \mathcal{D}_{1:n-1}) \leqslant \frac{1}{2} \sum_{t=0}^{T-1} \sum_{j=1}^{d_x} \log\left(1 + \frac{\sigma^2_{n-1,j}(z_t)}{\sigma^2}\right) \tag{2}$$

Tractable Exploration Objective: Maximize the upper bound

$$\pi^*_n = \arg\max_{\pi \in \Pi} \mathbb{E}_{\tau^\pi}\left[\sum_{t=0}^{T-1} \sum_{j=1}^{d_x} \underbrace{\frac{1}{2} \log\left(1 + \frac{\sigma^2_{n-1,j}(x_t, \pi(x_t))}{\sigma^2}\right)}_{r_{\text{info}}(x_t, \pi(x_t))}\right] \tag{3}$$

subject to: $x_{t+1} = f^*(x_t, \pi(x_t)) + w_t$

Interpretation

Collect data where model is most uncertain $\Rightarrow$ maximal information gain

Step 3: The Optimism Challenge ben

Problem: The planning objective requires knowing f^*, but f^* is unknown!

Naive Solution: Use mean estimate μ_{n-1} instead of f^*

$$x_{t+1} = \mu_{n-1}(x_t, \pi(x_t)) + w_t \tag{4}$$

Issue: Mean planner may believe uncertain regions are unreachable

OPAX Solution: **Optimism in the face of uncertainty**
Define confidence set of plausible dynamics:

$$\mathcal{F}_{n-1} = \left\{ \tilde{f} \mid \|\tilde{f}(z) - \mu_{n-1}(z)\| \leqslant \beta_{n-1}\sigma_{n-1}(z)\, \forall z \right\} \tag{5}$$

Key Insight: Plan assuming the **most optimistic** dynamics for reaching uncertain states

Step 4: Optimistic Planning Formulation ben

OPAX Optimistic Objective:

$$\pi_n, \eta_n = \arg\max_{\pi \in \Pi, \eta \in \Xi} \mathbb{E}_{\tau^{\pi,\eta}} \left[\sum_{t=0}^{T-1} \sum_{j=1}^{d_x} \log \left(1 + \frac{\sigma^2_{n-1,j}(\tilde{x}_t, \pi(\tilde{x}_t))}{\sigma^2} \right) \right] \tag{6}$$

subject to:

$$\tilde{x}_{t+1} = \mu_{n-1}(\tilde{x}_t, \pi(\tilde{x}_t)) + \beta_{n-1}\sigma_{n-1}(\tilde{x}_t, \pi(\tilde{x}_t))\eta(\tilde{x}_t) + w_t \tag{7}$$

where $\Xi = \{\eta : \mathcal{X} \to [-1, 1]^{d_x}\}$ is the space of **hallucination policies**

- $\eta(\tilde{x}_t)$ picks the most optimistic transition within confidence bounds
- Equivalent to $\max_{\tilde{f} \in \mathcal{F}_{n-1}} \mathbb{E}_{\tau^{\pi}|\tilde{f}}$[info reward]
- Guarantees $J_n(\pi^*) \leqslant J_n(\pi_n, \eta_n)$ w.h.p. (optimistic estimate)

Implementation

Solve as optimal control with extended action space (π, η) using MPC or SAC

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Assumptions

Regularity of the Dynamics

$f^\star$ lies in a RKHS with kernel k and $\|f^\star\| \leqslant 1$.

Recall

$$\gamma_n(k) = \max_{\mathcal{D}_1,\ldots,\mathcal{D}_n} \log\det(I + K_n)$$

and

$$\gamma_n(\text{linear}), \gamma_n(\text{RBF}) \asymp \text{polylog}(nT).$$

N. Srinivas et al. "Gaussian process optimization in the bandit setting: No regret and experimental design." ICML 2010. (The 1st presentation today!)

Main Guarantee

Theorem (Informal)

If we use $\mathcal{GP}(0, k)$ to build well-calibrated confidence intervals, then w.h.p.

$$\mathsf{E}_\pi\left[\underbrace{\max_t \|\sigma_n(s_t, a_t)\|_2^2}_{\text{maximum uncertainty along the trajectory}}\right] \leqslant \textit{poly}(T) \cdot \frac{\gamma_n(k)}{\sqrt{n}}, \quad \forall \pi,\ \forall n \in \mathbb{N}.$$

For linear and RBF kernels,

$$\lim_{n\to\infty} \sigma_n(s, a) \to 0.$$

Zero-Shot Performance

Corollary (Informal)

Consider an RL problem with a known bounded reward:

$$\arg\max_{\pi} R(\pi), \quad s_{t+1} = f^\star(s_t, a_t) + w_t.$$

If $n \gtrsim \text{poly}(d, T, \varepsilon^{-1}, \gamma_n(k))$, *then*

$$\max_{\pi} R(\pi) - R(\hat{\pi}) \leqslant \varepsilon,$$

where

$$\hat{\pi} \in \arg\max_{\pi} \min_{\eta} R(\pi, \eta), \quad s_{t+1} = \mu_n(s_t, a_t) + \eta(s_n)\beta_n\sigma_n(s_t, a_t) + w_t.$$

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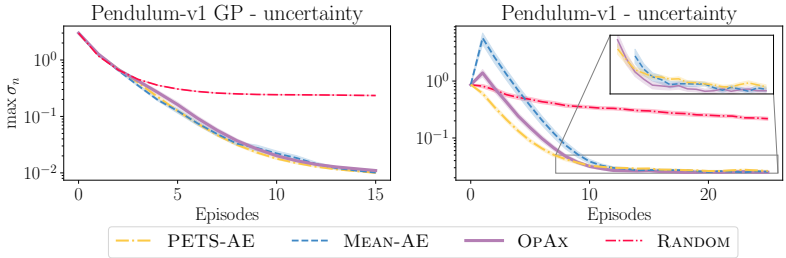
Experiments: Reduction in Epistemic Uncertainty (ok) ben

Does OPAX achieve its goal of reducing uncertainty everywhere?



Key Findings:

- OPAX reduces uncertainty much than random
- All active methods outperform random baseline
- Validates theoretical convergence guarantees

Implications:

- Information-driven exploration works in practice
- Both GP and PE models benefit from active exploration
- Optimism provides slight edge over mean planning

Experimental Setup and Baselines ben

Test Environments:

- Pendulum, MountainCar
- Reacher, Swimmer, Cheetah
- Fetch Manipulation (58D)

Key Comparisons:

- **OPAX** vs **H-UCRL** (task-specific SOTA)
- **OPAX** vs other active exploration methods

Evaluation Protocol:

1. Train models with different exploration strategies
2. Test on both **seen** and **unseen** tasks
3. Measure zero-shot performance

Models: GP (theory) & PE (scale)
Planning: MPC or SAC

Experiments: Learning and Zero-Shot Generalization ben

- `OpAx` performs **on par** with `H-UCRL` on trained tasks
- `OpAx` **outperforms** `H-UCRL` on some unseen tasks

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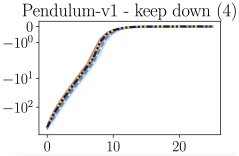
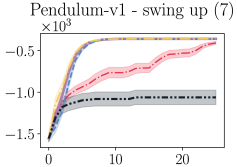
Evaluation (■: `H-UCRL`, ■: `OpAx`, ■: Random):



(a) Trained task



(b) Zero-shot task

Zero-shot swing up task:

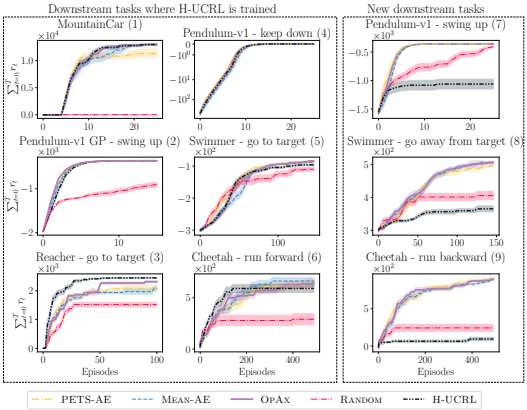
(play)

(c) `H-UCRL`

(play)

(d) `OpAx`

Experiments: Comprehensive Downstream Task Performance ben

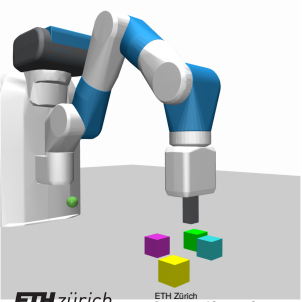


- **OPAX** consistently performs well across all environments
- On par with **H-UCRL** on trained tasks (1-6)
- **Outperforms H-UCRL** on unseen/new tasks (7-9) by a large margin
- **Random** baseline struggles, especially on MountainCar

Experiments: High-Dimensional Object Manipulation ben

Fetch Pick & Place Construction

- 7-DoF robot arm
- Four 6-DoF blocks
- 58-dim state space
- 4-dim actions

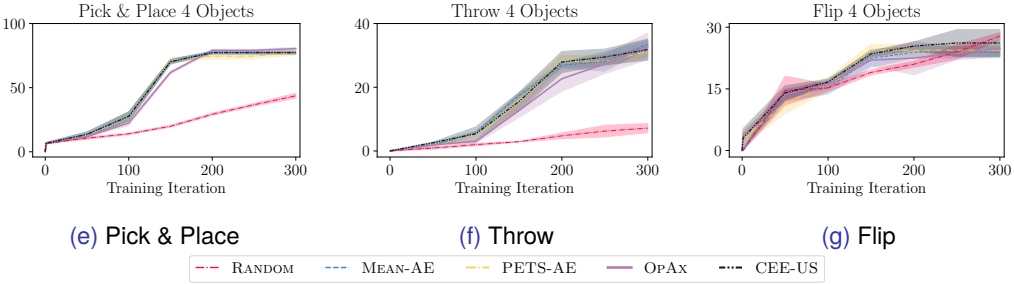


Zero-shot performance on three challenging tasks:

- Pick & Place, Throw, and Flip tasks
- OPAX matches or exceeds state-of-the-art baselines
- Comparison includes CEEUS (previous SOTA for active exploration)

Key finding: OPAX scales effectively to high-dimensional continuous control problems while maintaining strong zero-shot generalization

Experiments: Fetch Pick & Place Construction Results ben



(e) Pick & Place (f) Throw (g) Flip

- **OPAX** performs on par with or better than all baselines including CEEUS (state-of-the-art)
- Success rates evaluated via zero-shot planning with learned models
- Demonstrates scalability to high-dimensional (58D) continuous control

Key Findings ben

- **Sample Efficiency:**
  - OPAX often reaches target performance with fewer environment interactions
  - Active exploration methods significantly outperform random exploration
- **Generalization:**
  - OPAX matches H-UCRL on trained tasks
  - Significantly outperforms H-UCRL on unseen tasks (e.g., Cheetah run backward, Swimmer go away)
  - Task-specific methods fail to explore regions irrelevant to their training reward
- **Scalability:**
  - Successful application to high-dimensional Fetch task (58D) using PEs and MPC
  - Performance on par with CEEUS (previous state-of-the-art for active exploration)
- **Objective Robustness:**
  - Log-uncertainty objective (Eq. 6) performs similarly to simpler sum-of-uncertainties (OPAX-SUM)
  - Core benefit comes from optimistic planning rather than specific reward formulation

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Key Contributions and Impact ben

What OPAX Achieves:

- **First** practical algorithm with theory for continuous nonlinear dynamics
- Strong zero-shot generalization to unseen tasks
- Scales to high-dimensional problems (58D)
- Matches/exceeds SOTA baselines

Why It Matters:

- Moves beyond task-specific learning
- Enables true multi-task capabilities
- Foundation for continual learning agents
- Step towards general-purpose AI

Core Innovation

Applying **optimism** to **information gain** rather than rewards creates a principled exploration strategy that builds comprehensive world models

Open Challenges and Future Work ben

Current Limitations:

- Computationally expensive optimistic planning
- Requires well-calibrated uncertainty
- Limited to fully observed MDPs
- Pure exploration (no task integration)

Exciting Directions:

- Amortized optimistic planning
- Robust uncertainty estimation
- Extension to POMDPs
- Exploration-exploitation balance

Take-Home Message

OPAX demonstrates that **information-driven exploration** with **optimistic planning** can efficiently learn accurate, generalizable dynamics models – a crucial step towards truly adaptive AI systems.

References I

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